Example : 1. Tossing a coin. 2. Rolling a die. 3. Selecting a card from a pack of 52 cards.

Sample space

The set of all possible outcomes in a random experiment is called a sample space. It is generally denoted by S.

Example : When we roll a die, the possible outcomes are the face numbers 1,2,3,4,5,6 of the die. Therefore the sample space is $S = \{1,2,3,4,5,6\}$

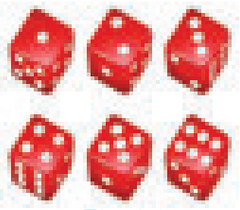


Fig. 8.2

Sample point Each element of a sample space is called a sample point.

8.4.1 Tree diagram

Tree diagram allow us to see visually all possible outcomes of an random experiment. Each branch in a tree diagram represent a possible outcome.

Illustration

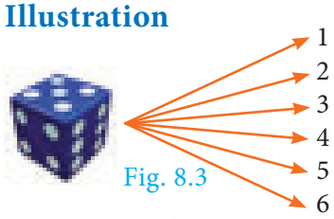


Fig. 8.3

(i) When we throw a die, then from the tree diagram (Fig.8.3), the sample space can be written as $S = \{1,2,3,4,5,6\}$

(ii) When we toss two coins, then from the tree diagram (Fig.8.4), the sample space can be written as $S=\{HH, HT, TH, TT\}$

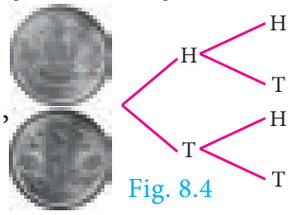


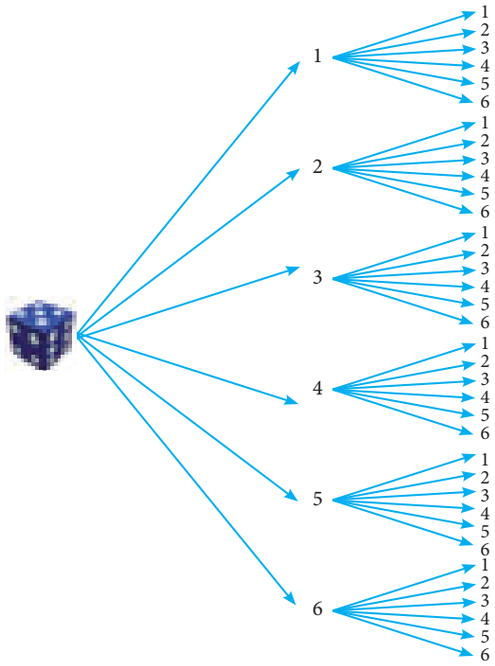
Fig. 8.4

Progress Check

1. An experiment in which a particular outcome cannot be predicted is called _____.
2. The set of all possible outcomes is called _____.

Example 8.17 Express the sample space for rolling two dice using tree diagram.

Solution When we roll two dice, since each die contain 6 faces marked with 1,2,3,4,5,6 the tree diagram will look like



Hence, the sample space can be written as

$S= \{(1,1),(1,2),(1,3),(1,4),(1,5),(1,6)$

$(2,1),(2,2),(2,3),(2,4),(2,5),(2,6)$

$(3,1),(3,2),(3,3),(3,4),(3,5),(3,6)$

$(4,1),(4,2),(4,3),(4,4),(4,5),(4,6)$

$(5,1),(5,2),(5,3),(5,4),(5,5),(5,6)$

$(6,1),(6,2),(6,3),(6,4),(6,5),(6,6)\}$

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